NumPy Statistical Analysis

1. Abstract

This project focuses on analyzing numerical data using Python and the NumPy library. The program calculates important descriptive statistics, including mean, median, standard deviation, minimum, maximum, quartiles and interquartile range. It also compares the mean and median to understand the influence of extreme observations and identifies potential outliers using the IQR method.

The project demonstrates how numerical analysis can help summarize datasets, understand data distribution and identify unusual values. These techniques form an important foundation for data science and machine learning.

2. Introduction

Statistical analysis is an essential part of data analysis because it helps transform raw numerical observations into meaningful information. Measures such as mean, median and standard deviation help describe the central tendency and variability of data.

Python provides several libraries for numerical computation. NumPy is particularly useful because it offers efficient array operations and built-in statistical functions.

In this project, a numerical dataset is analyzed using NumPy. The results are displayed in a readable format to make the statistical properties of the dataset easier to understand.

3. Objectives

The primary objectives are:

- To create and analyze a numerical dataset.
- To calculate mean and median.
- To determine population standard deviation.
- To identify minimum and maximum values.
- To calculate the first and third quartiles.
- To calculate the interquartile range.
- To identify potential outliers.
- To compare mean and median and interpret the results.

4. Tools and Technologies

Python: Used to implement the statistical analysis program.

NumPy: Used to perform numerical calculations and statistical operations.

CSV: Used to store the numerical observations in a structured format.

Visual Studio Code / Jupyter Notebook: Used to write, execute and test the program.

5. Dataset Description

The project uses a sample dataset containing 15 numerical observations.

The values are:

45, 50, 52, 55, 58, 60, 62, 65, 68, 70, 72, 75, 78, 80, 150

Most observations range between 45 and 80. The value 150 is deliberately included as a high observation to demonstrate its effect on descriptive statistics and potential outlier detection.

The dataset is stored in `dataset.csv`, with a numerical column named `value`.

6. Methodology

The project follows these steps:

1. Load the dataset from the CSV file.
2. Read the numerical observations into a NumPy array.
3. Validate the dataset to ensure that it contains at least three finite numerical values.
4. Calculate the mean and median.
5. Calculate the population standard deviation.
6. Identify the minimum and maximum values.
7. Calculate the first quartile (Q1) and third quartile (Q3).
8. Calculate the interquartile range (IQR).
9. Calculate the lower and upper outlier fences.
10. Compare the mean and median and display an interpretation.

7. Statistical Concepts and Formulas

7.1 Mean

The mean is the arithmetic average of all observations.

Formula:

Mean = Sum of all observations / Number of observations

The mean provides a measure of central tendency but can be influenced by extreme values.

7.2 Median

The median is the middle value when observations are arranged in ascending order.

When the number of observations is even, the median is the average of the two middle values.

The median is less sensitive to extreme values than the mean.

7.3 Standard Deviation

Standard deviation measures the spread of observations around the mean.

For population standard deviation:

$$\sigma = \sqrt{[\Sigma(x_i - \mu)^2 / N]}$$

Where:

- x_i represents an observation.
- μ represents the population mean.
- N represents the total number of observations.

The program uses NumPy's `np.std()` function with its default population convention (`ddof=0`).

7.4 Minimum and Maximum

The minimum is the smallest observation, while the maximum is the largest observation.

These values indicate the range covered by the dataset.

7.5 Quartiles and Interquartile Range

Q1 represents the 25th percentile, and Q3 represents the 75th percentile.

Formula:

$$IQR = Q3 - Q1$$

The IQR measures the spread of the middle 50% of the observations.

7.6 Potential Outlier Detection

The IQR method uses the following fences:

$$\text{Lower fence} = Q1 - 1.5 \times IQR$$

$$\text{Upper fence} = Q3 + 1.5 \times IQR$$

Values below the lower fence or above the upper fence are flagged as potential outliers. An outlier is not automatically an error; it may represent a genuine observation.

8. Implementation

The project is implemented in Python using NumPy and the built-in CSV module.

The main program performs the following operations:

- Reads numerical data from `dataset.csv`.
- Converts observations into a NumPy array.
- Calculates descriptive statistics.
- Calculates quartiles and the IQR.
- Detects potential outliers using the IQR fences.
- Compares the mean and median.
- Prints the results in a structured terminal output.

The main source code is stored in `main.py`.

9. Results and Analysis

The program analyzes 15 observations and displays the calculated statistics.

The expected results for the supplied dataset are approximately:

Statistical Measure	Result
Number of observations	15
Mean	68.00
Median	65.00
Population standard deviation	23.46
Minimum	45.00
Maximum	150.00
First quartile (Q1)	56.50
Third quartile (Q3)	76.50
Interquartile range	20.00
Lower fence	26.50
Upper fence	106.50
Potential outlier	150.00

The mean is higher than the median, indicating that the unusually high observation pulls the arithmetic average upward.

The IQR method flags 150 as a potential outlier because it exceeds the upper fence of 106.50.

The results illustrate why it is useful to examine multiple statistical measures instead of relying on the mean alone.

10. Applications

The statistical techniques used in this project are applicable to:

- Student marks and academic performance analysis.
- Sales and revenue analysis.
- Employee salary analysis.
- Exploratory data analysis.
- Data quality checks.
- Identifying unusual numerical observations before machine learning.

11. Limitations

The project uses a small, manually created dataset. Its findings may not represent a larger real-world population.

Outlier detection using the IQR method depends on the distribution of the data and should be interpreted alongside domain knowledge. A potential outlier is not necessarily incorrect.

The program analyzes a single numerical column and does not currently perform multivariable analysis or generate visualizations automatically.

12. Future Enhancements

The project can be extended by:

- Allowing users to upload their own CSV datasets.
- Adding histograms and box plots.
- Supporting multiple numerical columns.
- Exporting statistical summaries to CSV or PDF.
- Creating a simple interactive dashboard.
- Comparing statistical measures across different groups.

13. Conclusion

This project demonstrates how Python and NumPy can be used to perform basic statistical analysis efficiently. It calculates measures of central tendency, variability and data spread, and uses the IQR method to identify potential outliers.

The project strengthens practical understanding of numerical data analysis and provides a foundation for more advanced exploratory data analysis and machine learning workflows.

14. References

1. NumPy Official Documentation: https://numpy.org/doc/
2. Python Official Documentation: https://docs.python.org/3/
3. NumPy Statistical Functions: https://numpy.org/doc/stable/reference/routines.statistics.html

15. Project Evidence

GitHub repository: https://github.com/venky1318/numpy-statistical-analysis